

B.E. I YEAR EXAMINATION JAN 2024

IT-A, IT-B and All Ex Students

1ACRC2: Applied Chemistry & Environment Science

Duration: 3 Hrs.

Max. Marks : 60

Note: Attempt any two parts from every question. All questions carry equal marks. Answer must be brief and to the point. Any assumptions made answering the questions should be stated. Give chemical equations and diagrams wherever needed. Avoid use of additional answer sheets.

- Q. 1**
- (a) Write descriptive note on corrosion of boiler.
 - (b) What is EDTA? Explain basic principle of EDTA method. 0.25 g of CaCO_3 was dissolved in HCl and the solution was made to one liter with distilled water. A total of 100 mL of the above solution required 25 mL of EDTA solution on titration. 100 mL of the hard water sample required 35 mL of the same EDTA solution on titration. After boiling 100 mL of this water, cooling, filtering and then titration required 10 mL of EDTA solution. Calculate the temporary and permanent hardness of water.
 - (c) Explain chemistry and calculate the amount of lime and soda needed for softening 1,00,000 liters of water containing the following:
 $\text{HCl} = 8.9 \text{ mg/L}$; $\text{Al}_2(\text{SO}_4)_3 = 38.9 \text{ mg/L}$; $\text{MgCl}_2 = 10.2 \text{ mg/L}$; $\text{NaCl} = 30.38 \text{ mg/L}$
- Q. 2**
- (a) Discuss various important properties of refractories?
 - (b) Describe industrial manufacturing process of Portland cement with their chemistry.
 - (c) Explain mechanism of polymerization with suitable examples.
- Q. 3**
- (a) What is lubrication? Discuss function and classification of lubricants.
 - (b) What do you mean by blended or compounded oil? What are the various additives used to impart desirable properties to lubricants?
 - (c) What is flash and fire point? Discuss its significance and process of determination.
- Q. 4**
- (a) Define chemical shift in NMR? Giving suitable examples explain factors affecting the position of chemical shift.
 - (b) Discuss basic principle behind IR absorption by molecules. Calculate stretching vibrational frequencies for $\text{C}=\text{C}$ and $\text{C}-\text{H}$ bonds in ethylene. (Assume approximate force constants).
 - (c) Write explanatory note on construction, working and applications of colorimeter.
- Q. 5**
- (a) Describe air pollution including sources of pollutants, harmful effects and control measures.
 - (b) Discuss atmosphere and its components.
 - (c) How does water pollution impact ecosystems and human health, and what measures can be taken to mitigate its effects?